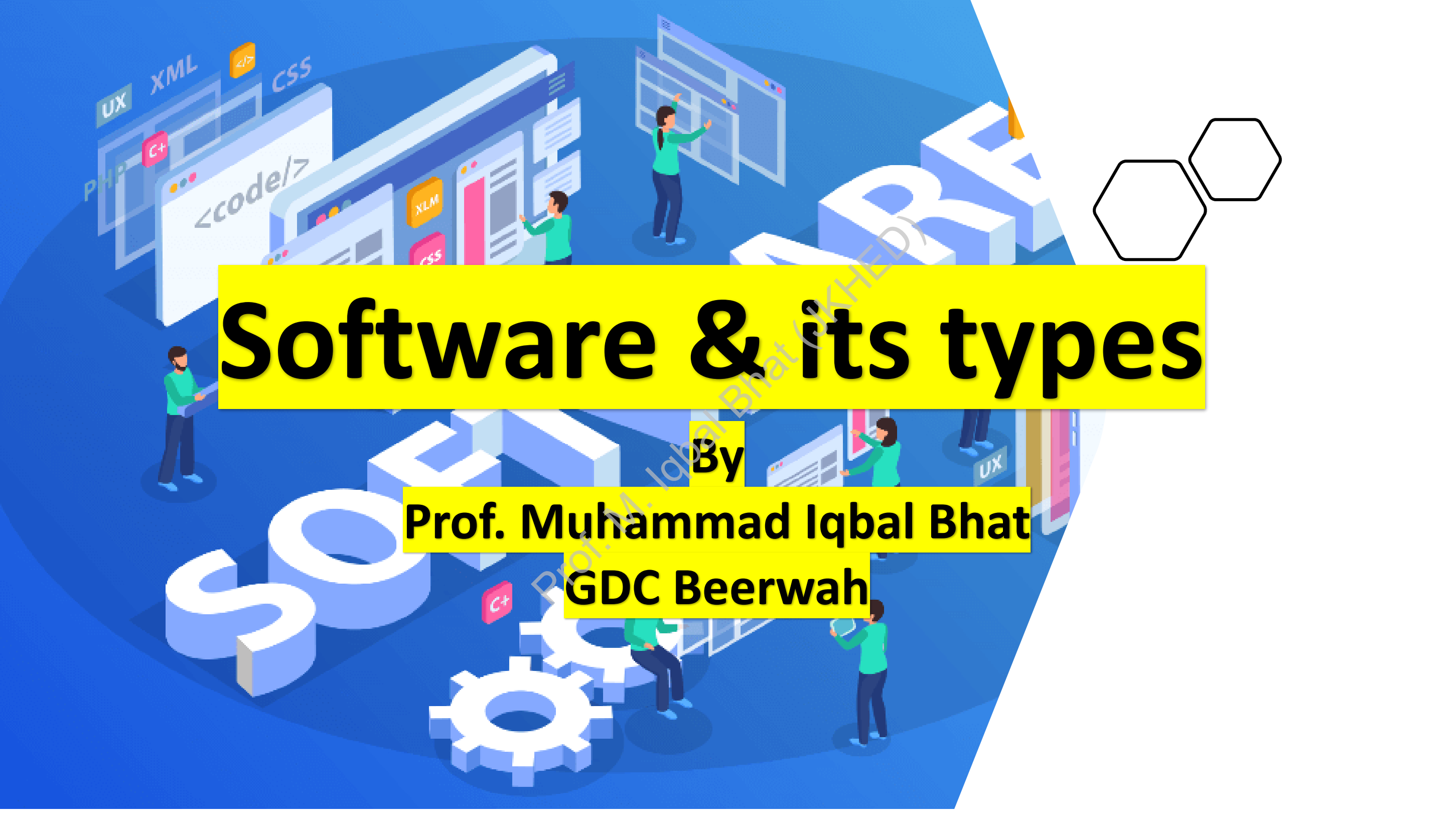


An isometric illustration on a blue background depicting software development. It features large 3D letters for 'SOFTWARE' and 'WEB'. Various icons represent programming languages and technologies: 'UX', 'XML', 'CSS', 'PHP', 'C+', '<code/>', 'XLM', and 'CSS'. Several stylized human figures are shown interacting with these elements, such as pointing at screens or sitting on gears. In the top right corner, there are three hexagons: two white with black outlines and one solid black.

Software & its types

By

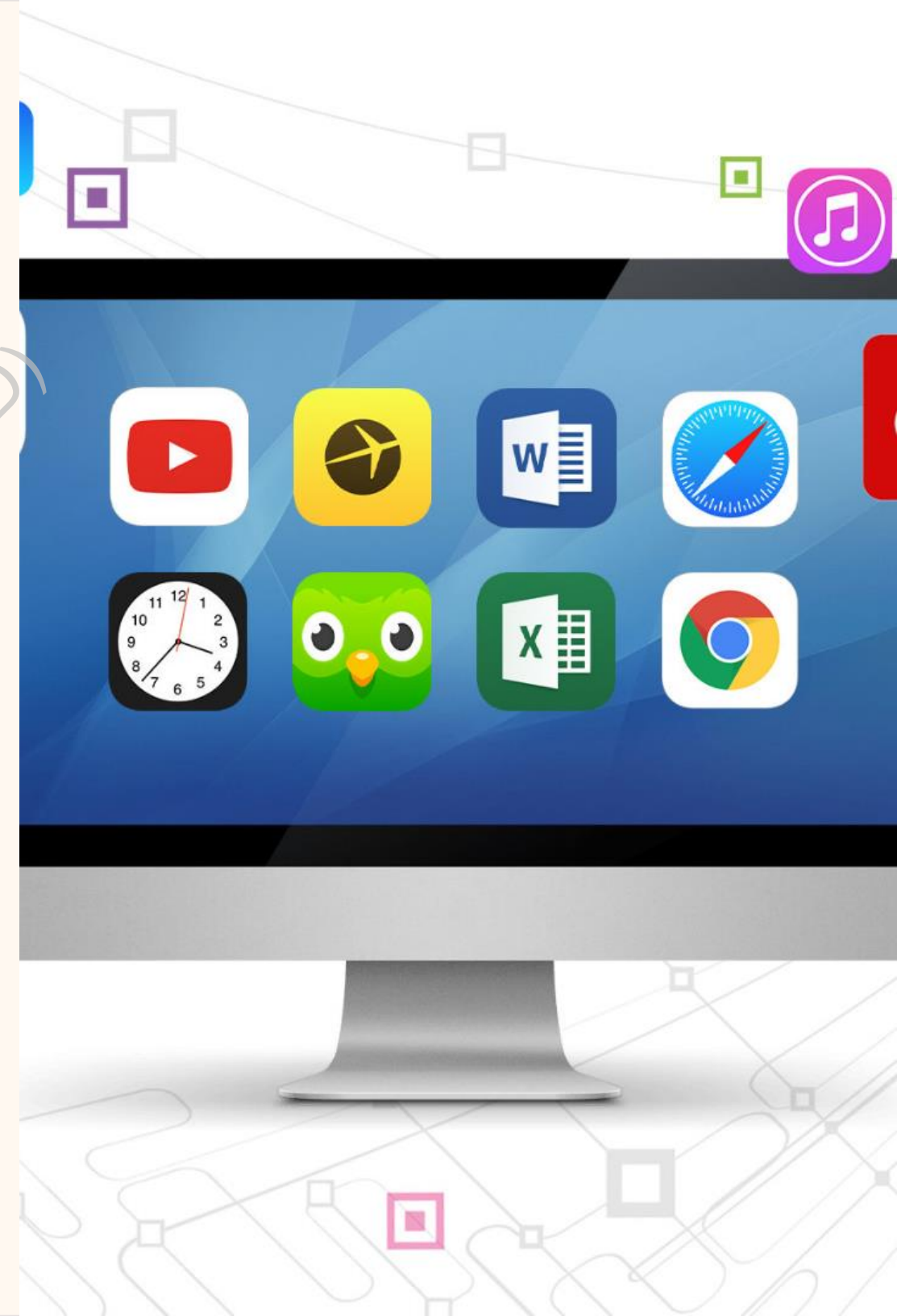
Prof. Muhammad Iqbal Bhat
GDC Beerwah

Software and its types

Software plays a crucial role in our digital world. Let's explore the different types and understand their significance in various applications.



by Prof. Mohd Iqbal Bhat



Software

Software is a set of instructions, data, or programs used to operate a computer and execute specific tasks.





Software

Hardware

Definition	Collection of instructions that enables a user to interact with the computer	Physical devices that are required to store and execute (run) the software
Types	System software and application software	Input, storage, processing, control, and output devices
Function	Provides the instruction to the hardware	Performs the task at machine level
Faults	Software failure is systematic. Software does not have an increasing failure rate.	Hardware failure is random. Hardware has increasing failure at the last stage.
Durability	Software is durable and doesn't wear out, but with time, <u>bugs</u> may arise in the software which could be rectified.	Hardware wears out with time.
Nature	Software is logical in nature.	Hardware is physical in nature.
Examples	QuickBooks, Adobe Acrobat, <u>Google Chrome</u> , Microsoft Word, Microsoft Excel, Apple Maps	CPU, <u>Hard drive</u> , <u>RAM</u> , keyboard, mouse, <u>USB drive</u>


```
self.connection._  
if self._check_packet_is_eo  
self.unbuffered_active  
self.connection = None  
self.rows = None  
return
```

```
row = self._read_row_from_p  
self.affected_rows = 1  
self.rows = (row,) # rows  
return row
```

```
finish_unbuffered_query(se
```

After much reading on the
in fact, no way to stop M
executing a query, so we

Open Source vs Proprietary Software

Prof. M. Iqbal Bhat (JKUED)

Open Source Software

Open source software refers to computer programs and software applications for which the original source code is made freely available to the public.

Accessibility

Open source software allows users to access and modify the source code freely.

Community

It encourages collaboration and a supportive community of developers and contributors.

Customization

Users have the freedom to tailor the software to their specific needs and preferences.

Transparency

The source code is visible, providing transparency and enabling trust for users.



Advantages of Open Source Software

1 Cost Savings 💰

No licensing fees or expensive subscriptions, making it budget-friendly.

2 Flexibility 🧘

Open source software can be adapted and integrated easily with existing systems.

3 Strong Security 🔒

With a large community of developers, vulnerabilities are quickly identified and fixed.

4 Continuous Improvement 🚀

Regular updates and contributions ensure a constantly evolving and improving product.

Disadvantages of Open Source Software

1 Learning Curve

Open source software may require technical expertise to understand and utilize effectively.

2 Limited Support

Support options may be limited compared to proprietary solutions.

3 Compatibility Challenges

Some open source software may have compatibility issues with certain systems or hardware.

4 Fragmentation

Multiple versions and distributions can lead to fragmentation and inconsistency.



Proprietary Software

Proprietary software, also known as closed-source software, is a type of computer software for which the source code is not made available to the public. In contrast to open source software, proprietary software is typically owned by a company or individual who retains control over the software and its distribution.

1 Ownership

Proprietary software is owned and controlled by a company or individual.

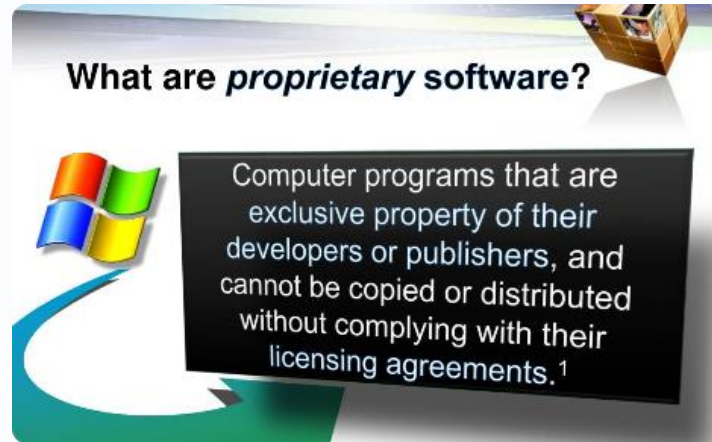
2 Licensing

Access to the source code and usage is restricted by licenses.

3 Trade Secrets

Companies can protect their code and maintain a competitive advantage.

Advantages of Proprietary Software



Customer Support ☎

Proper customer support ensures a smooth user experience and assistance when needed.



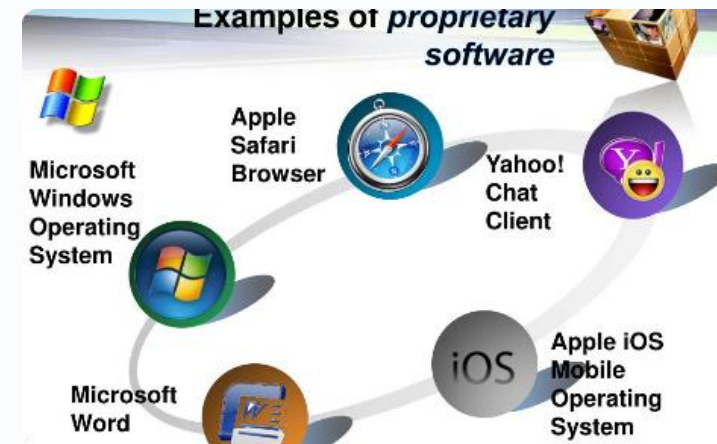
Seamless Integration 🧩

Proprietary solutions are often designed with specific systems in mind, ensuring smooth integration.



Innovation 🚀

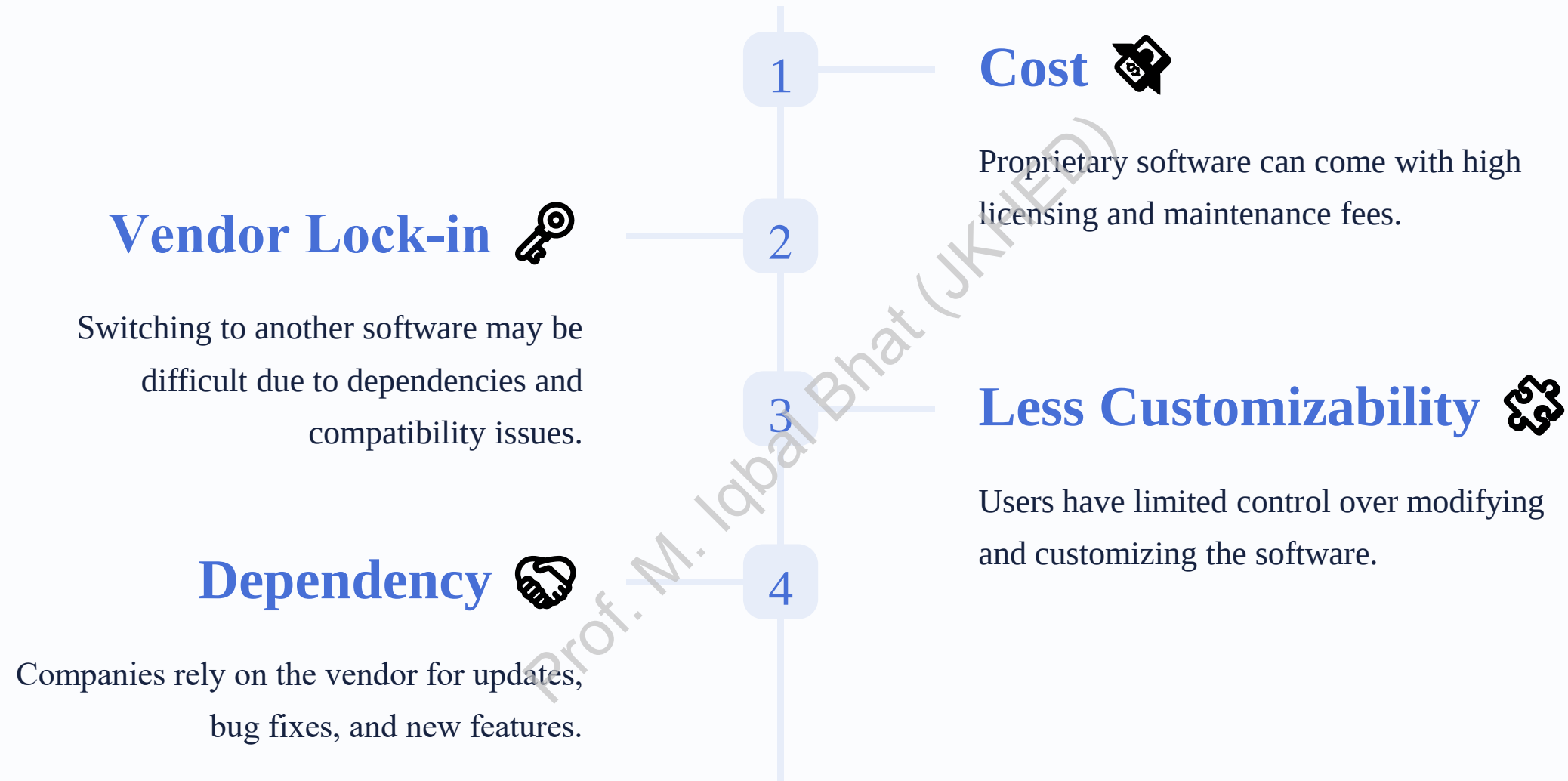
Companies have more control over innovation and can invest in research and development.



Enhanced Security 🔒

Proprietary software often provides robust security measures to protect sensitive data.

Disadvantages of Proprietary Software

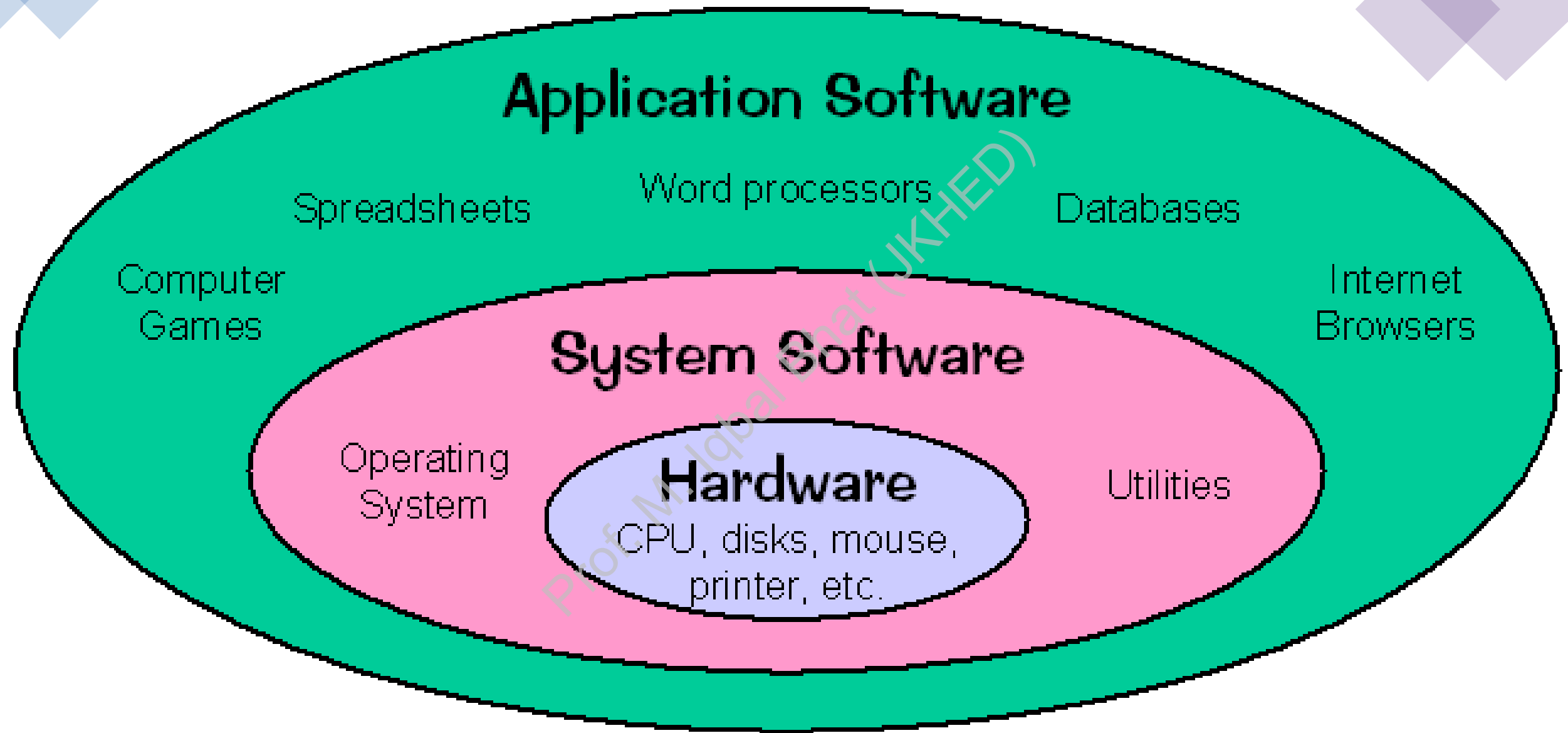


Types of Software

- System Software
- Application Software
- Programming Software

Software can be broadly categorized into system, application, and programming software. Each type serves a specific purpose in the digital ecosystem.

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Application Software

- Application software is software that helps an end-user complete tasks such as doing research, taking notes, setting an alarm, designing graphics, or keeping an account log.
- Application software lies above the system software and is different from system software in that it's designed for the end use and is specific in its functionality.
- This type of software is sometimes referred to as non-essential software because it's installed and operated based on the user's needs.
- Any application on a mobile phone is an example of application software.



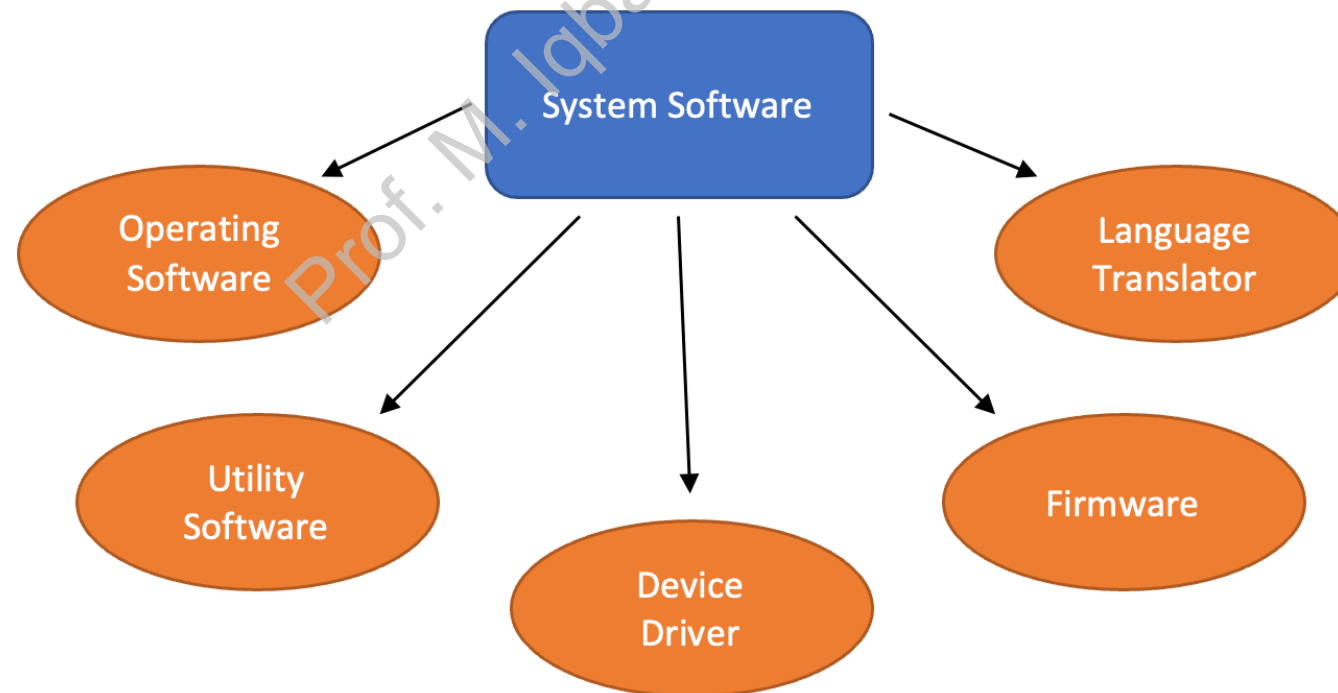
Application Software



- **Word processors:** Applications used for documentation. Examples include Microsoft Word, Google Docs, and AppleWorks
- **Spreadsheet software:** Software used to compute quantitative data. Examples include Microsoft Excel, Google Sheets, and Quattro Pro
- **Database software:** Software used to create and manage a database to organize data. This is also known as database management software (DBMS). Examples include MySQL, Clipper, and FileMaker
- **Multimedia software:** Tools that are able play, create, or record images, audio, or video files. It's used for video editing, animation, graphics, and image editing. Examples include Adobe Photoshop and Picasa
- **Application suites:** A collection of related programs sold as a package. Microsoft Office is the most widely used application suite.
- **Internet browsers:** Software used to access and view websites. Examples include Google Chrome and Internet Explorer.
- **Email programs:** Software used for emailing. Examples include Outlook and Gmail.

System Software

- Software required to run the hardware parts of the computer and other application software are called system software.
- System software acts as an interface between hardware and user applications. An interface is needed because hardware devices or machines and humans speak in different languages.
- It's essential in managing the whole computer system - [?]
- When a computer is first turned on, it's the system software that is initially loaded into memory.
- Unlike application software, the system software isn't directly used by end users. Instead, it runs in the background of a device.



Operating System

- System software that is responsible for functioning of all hardware parts and their interoperability to carry out tasks successfully is called operating system (OS).
- OS is the first software to be loaded into computer memory when the computer is switched on and this is called booting.
- OS manages a computer's basic functions like storing data in memory, retrieving files from storage devices, scheduling tasks based on priority, etc.



Android



Windows 10



MICROSOFT
Microsoft
Windows



ibuntu
Ubuntu



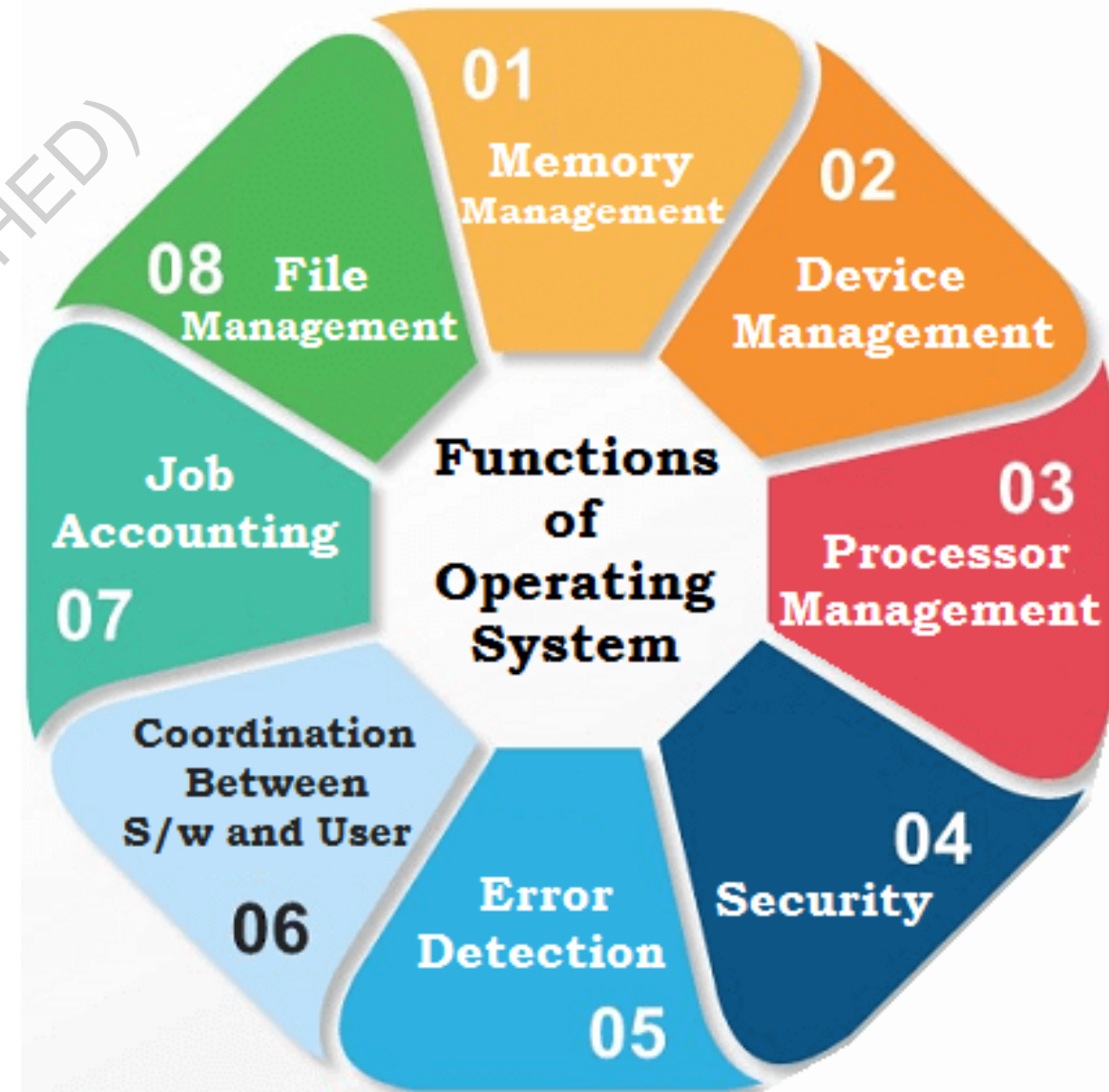
Linux



macOS



Chrome OS



Operating System



Cost

Free of charge or lower cost

Must pay to use or higher cost



Source Code

With source code

Without source code



Flexibility

Can modify software

Cannot modify software



Installation

Can install freely

License required



Support

Rely on loyal and engaged online user community to deliver support

Full support from vendor if any problem occurs

Operating System

- **Singleuser operating system**
 - It allows only one users to use same computer at same time e.g. DOS.
- **Multiuser operating system**
 - It allows multiple users to use same computer at same time e.g. Linux, Unix and Windows server 2008.
- **Multiprocessor operating system**
 - It supports two or more processors running programs at the same time e.g. Linux, Unix and Windows server 2008.

Operating System

- **Multitasking operating system**

- It can execute more than one programs at the same time e.g. Unix and windows 8.1

- **Time sharing operating system**

- It allows many users to share the computer simultaneously. It is used when several users are linked through communication networks to a single computer.

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Types of operating systems

- Different types of operating system are as follows
 - Stand alone operating systems
 - Server operating systems
 - Mobile operating systems

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Types of operating systems

- ***A stand alone operating system*** works on a desktop or laptop computer.
- Some stand alone operating systems can work with a server operating system and are called client operating system.
- Some examples are DOS, Windows, Unix, Linux and Mac OS.
- Windows is the most widely used OS developed by Microsoft.

Types of operating systems

- ***A server operating system*** is an operating system that is designed to support a network.
- It is also called network operating system.
- It usually resides on a server and client computers on the network depend on the server for resources.
- Some examples are Windows Server 2012/2016/2019, OS X Server, Sun Solaris, NetWare,

Types of operating systems

- ***Mobile operating system*** is an operating system that is used in handheld computers and mobile devices.
- It provides graphical user interface (GUI) and other features such as touch screen support, navigation systems, speech recognition, wireless connectivity etc.
- Some examples are Android, iOS, Windows phone and BlackBerry.

Utility Programs

- ***Utility program*** is a software that is used for effective management of computer system.
- It keeps the computer system running smoothly.
- Users can use utility programs to perform maintenance tasks related to different devices and programs.
- Most operating systems include different built-in utility programs.

Types of Utility Programs

- Different types of utility programs are
- **Antivirus**
 - It is an application that is certain to be installed on every operating system, especially in the operating system Windows. Because without Antivirus, the operating system will be very high risk of loss or damage to a number of data and theft of sensitive data can occur with all hacking techniques.
 - Examples of popular antivirus applications such as Avast, SmAdav, AVG, BitDevender, and so forth. The variants of antivirus are also varied, some are paid and some are free.

Types of Utility Programs

- **Archivers and Data Compression**

- Software that can make a file into an archive and can also reduce or compress its size.
- An example is when you have 80 pdf files that you will send via email to your co-workers. If you want to send one file at a time, as many as 80 pdf files through attachment tools in the email, then that will be very troublesome for you. You can send as many as 80 pdf document files by archiving them in a .zip file and sending them via email.
- Some examples are WinZip, WinRAR, 7-Zip, FilZip

Types of Utility Programs

- **Backup Software**

- If the WinZip and FilZip software as above can be used to create an archive of a file, then the backup software can be used to create an archive from the hard disk drive partition.
- That way it can also be said that the data to be archived is the entire partition. As for some software that can perform backup needs, such as Nova Backup software, Norton Ghost and others.

Types of Utility Programs

- **Recovery Software**

- Data is an asset that is very important for today's technological life. Indeed a number of data that you have deleted or when you do *quick form disk*, that means only the database table is left blank.
- Some of your actual old data is still in storage and has not been completely erased. So because of that, we can still restore power that has been erased by using data recovery software. Some data recovery software, such as Recuva, TestDisk, and so forth.

Types of Utility Programs

- **Software Uninstaller**

- You all must have uninstalled some of the software that is on your computer. By uninstalling the software, you no longer need the application. But have you ever found an application that cannot be uninstalled? If true, then you can overcome it with an uninstaller software.
- Some kinds of uninstaller software such as REVO uninstaller, IObit Uninstaller, Geek Uninstaller, and so on.

Types of Utility Programs

- **Disk Cleaner**

- Disk cleaner software works to clean the files on the hard disk drive that is no longer useful. Usually we mention it with junk files, that way, the free space on the hard disk drive can be optimized.
- An example of this Disk Cleaner application is CCleaner. Where the application has the main feature to clean disk files that are no longer useful.

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Types of Utility Programs

- **Personal firewall**

- It is used to detect and protect a personal computer from unauthorized intrusions.
- Built-in personal firewall is automatically enabled when windows is installed.

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Programming Software

Integrated Development Environment (IDE)

IDEs like Visual Studio or IntelliJ IDEA provide tools for writing, debugging, and testing code.

Text Editors

Lightweight editors like Sublime Text or Vim focus on code editing and customization.

Version Control

Tools like Git and SVN track changes and facilitate collaboration among developers.

Language Processors

- An important function of system software is to convert all user instructions into machine-understandable language.
- System software that converts source code to object code is called a language processor.



Language Processors

